

Acne Scar Management

Scarring is often a significant concern for patients, and the treatment approach should be determined by scar characteristics, the severity of scarring, and the area of involvement. Once scarring has developed, management considerations include the cost of treatment, severity of lesions, patient expectations, physician goals, and the psychological impact of the scars. Recently, Fife suggested key questions to guide clinicians during the acne scar history (Table 6). Additional tips for the physical examination include:

- Shining light across the skin to highlight atrophic areas
- Asking the patient to identify areas of concern using a mirror
- Assessing the physical characteristics of the scar (color, depth, width, size)
- Stretching the skin to determine whether the scar flattens or disappears

If hyperpigmentation is present, the patient may be at increased risk of transient post-inflammatory hyperpigmentation following acne scar treatments, particularly excisional or resurfacing procedures.

Treatment Considerations

Although a variety of scar treatments are available, there is currently no gold standard due to insufficient evidence supporting first-line use of any single intervention (Table 7). Furthermore, the relative safety profiles of various interventions have not been fully established. Not all patients respond to the same procedures, and a fast, permanent solution is rarely achievable. Often, a combination of treatment modalities is required.

Medical therapies can be combined with procedural interventions, and multiple procedures may be used sequentially. Some patients may prefer a series of gentler treatments with minimal downtime rather than a single aggressive procedure. Clinicians should help patients develop realistic expectations regarding potential improvements and ensure they understand the risks associated with each treatment option.

Categories of Acne Scar Treatments

There are several broad categories of procedures used in acne scar management:

Resurfacing Procedures

These aim to injure the epidermis and superficial dermis to stimulate neocollagenesis and promote epidermal remodeling.

- **Chemical peels**

- Full face

- CROSS technique (Chemical Reconstruction of Skin Scars)

- **Dermabrasion**

- **Laser resurfacing**

- Ablative

- Non-ablative

- Fractional

Lifting Procedures

These elevate the base of the scar to bring it closer to the level of surrounding skin.

- **Subcision**

- **Fillers**

- Directly under scars

- Volumizing

- Autologous fat transfer

- **Punch elevation**

Excisional Techniques

Used primarily for deep, sclerotic, or hypopigmented scars.

- **Punch excision**

- **Elliptical excision**

- **Punch grafting**

Other Techniques

- **Skin needling** (microneedling)
- **Facelift** (in select cases with significant tissue laxity)
- **Combination techniques** (tailored approaches using multiple modalities)

Risks and Special Considerations

All procedures carry potential risks, including infection, hyperpigmentation, prolonged erythema, and poor wound healing. These risks may be heightened in individuals with darker skin tones. Truncal acne scars are generally more challenging to treat than facial scars.

Table 6: Acne Scar History

Adapted with permission from Fife.

Current Acne Assessment

- Are you using an acne treatment now?

Patient-Specific Questions

- What aspect of your skin is most bothersome? (e.g., dark spots, acne, wrinkles, other)
- Please identify scars or areas of your face that bother you the most
- How do the scars affect your lifestyle?
- Do you have time constraints due to work or travel?

Questions That Could Affect the Therapeutic Regimen

- Have you done anything in the past to treat your scars?
- If yes: How many sessions? What was the downtime? How effective was it? Any healing problems?
- What do you want to achieve with treatment?
- Did you need isotretinoin to treat your acne? If yes, when was your last dose?
- Does your skin tend to darken after acne lesions, surgery, or other injury?
- Do you have any painful, thick, or itchy scars?

Table 7: Interventions for Treating Facial Atrophic Acne Scars

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Resurfacing Procedures

- Chemical peels
- Full face
- CROSS technique
- Dermabrasion
- Laser resurfacing
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- Non-ablative
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